

2 Steelhead trout, *Oncorhynchus mykiss*, are fish that live in streams in North America.

To increase the number of steelhead trout, captive breeding has occurred since 1992. Fish eggs and sperm are mixed and the young fish grow in large tanks of aerated water for the first year of their lives. Most are then released into the wild, however a few male and female fish are kept to become the parents of the next generation of captive-bred fish.

Each tank may hold up to 50 000 fish. The young captive fish are fed processed food. Some young fish are unable to survive these conditions and a proportion die. Death is usually the result of poor wound-healing after accidents due to overcrowding and due to the spread of diseases.

(a) (i) Name the expected pattern of variation in wound-healing ability in a population of fish.

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(ii) Name the process that results in improved survival of captive fish in second, third and subsequent generations of captive-bred fish.

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(b) Suggest **and** explain **three** ways in which the tank environment may make the phenotype of a captive fish different from a wild fish.

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(ii) Explain how gene expression is controlled in eukaryotes such as fish.

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(d) Many of the differences in gene expression between the offspring of the wild and captive-bred fish were in genes coding for proteins involved in the immune response and in genes coding for proteins involved in wound healing. The fish from captive-bred parents expressed these genes to a greater degree.

The researchers concluded that the differences were inherited and adaptive.

Explain why the researchers concluded that the differences in gene expression between the two groups of fish were:

(i) inherited

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(ii) adaptive.

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[Total: 16]